# IGCSE 0607 Practice Questions — E3.3 — Combined Questions & Markscheme

## Functions

**Total Questions:** 30 — **Total Marks:** 234 Marks

## Table of Contents & Quick Navigation


- **E3.3** Functions
  → **Paper 2 (Non-Calculator)** *(16 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

**1** Functions $f$ and $g$ are defined as follows.

$f(x) = 2x - 5$, $g(x) = x^2 + 1$

**(a)** Find the value of $f(g(3))$.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**2** It is given that $f(x) = 3x - 2$ and $g(x) = \dfrac{10}{x}$.

**(a)** Find $f(4)$.

..... **[1]**

**(b)** Find $x$ when $g(x) = 2$.

$x =$ ..... **[1]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**3** The functions $f$ and $g$ are defined as $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

**(a)** Work out

**(i)**

hspace*0.8cm $f(-4)$

..... **[1]**

**(ii)**

hspace*0.8cm $gf(2)$

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**4** $f(x) = 4x - 5$
$g(x) = \dfrac{x}{3} + 2$

**(a)** Work out $f(-2)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $fg(x)$ in its simplest form.

$fg(x) =$ ..... **[2]**

**5** The functions $f$ and $g$ are defined as $f(x) = 5x - 2$, $g(x) = x^2 + 3$.

**(a)** Find the value of $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $fg(2)$.

..... **[2]**

**6** Consider the functions $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

**(a)** Find $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $gf(2)$.

..... **[2]**

**7** Let $f(x) = 5 - 2x$, $g(x) = x^2 - 3$.

**(a)** Work out $f(-4)$.

..... **[1]**

**(b)** Work out $gf(2)$.

..... **[2]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**8** $f(x) = 4x - 3$, $g(x) = 2 - 5x$

**(a)** Work out $f(2)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Work out $gf(1)$.

..... **[2]**

**9** Let $f(x) = 4x - 3$ and $g(x) = 2x^2 + 1$.

**(a)** Work out $fg(3)$.

..... **[2]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Solve the equation

$$f(x) = 17$$

$x =$ ..... **[2]**

**10** Two functions, $f$ and $g$, are defined as follows:

$$f(x) = 3x - 2 \qquad g(x) = x^2 + 1$$

**(a)** Work out $f(4)$.

..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(c)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

**11** The functions $f$ and $g$ are defined as $f(x) = \dfrac{3x+2}{x-1}$, $x \neq 1$, $g(x) = x - 2$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[4]**

**12** Consider the functions $h(x) = \dfrac{12}{x-1}$ and $k(x) = 2x + 3$.

**(a)** State the value of $x$ which must be excluded from the domain of $h(x)$.

$x =$ ..... **[1]**

**(b)** Find $k^{-1}(x)$.

$k^{-1}(x) =$ ..... **[2]**

**(c)** Find $hk(x)$, giving your answer in its simplest form.

$hk(x) =$ ..... **[2]**

**(d)** Solve the equation.

$h(x) = k(2)$

$x =$ ..... **[3]**

**13** Let $f(x) = \dfrac{4}{2x-3}$ for $x \neq 1.5$, and $g(x) = x + 5$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Solve the equation $f(x) = g(-3)$.

$x =$ ..... **[3]**

**(c)** Express $gf(x)$ as a single fraction in its simplest form.

$gf(x) =$ ..... **[3]**

**14** The functions $f$ and $g$ are defined as $f(x) = \dfrac{x+3}{x-1}$, where $x \neq 1$, and $g(x) = 2x - 5$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Show that $ff(x) = x$.

**[3]**

**(c)** Solve the equation $g(x^2) = f(3)$.

$x =$ ..... **[4]**

**15** $f(x) = \dfrac{3x+2}{x-1}$, $g(x) = \dfrac{2}{x+1}$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Find $fg(x)$, giving your answer as a single fraction in its simplest form.

..... **[3]**

**(c)** Solve the equation.

$f(x) = f^{-1}(x)$

Give your answers in the form $a \pm \sqrt{b}$, where $a$ and $b$ are integers.

$x =$ ..... **[4]**

**16** The functions $f$ and $g$ are defined as follows:

$f(x) = \dfrac{3x+5}{x-1}, x \neq 1$, and $g(x) = x^2 + 1$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

**(b)** Solve the equation $f(x) = x$.

$x =$ ..... , $x =$ ..... **[4]**

**(c)** Find $gf(2)$.

$gf(2) =$ ..... **[2]**

**17** The functions $h$ and $k$ are defined as $h(x) = \dfrac{2x+1}{3}$ and $k(x) = 5 - x$.

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

**(b)** Find an expression for $hk(x)$ in its simplest form.

$hk(x) =$ ..... **[2]**

**(c)** Solve the equation $h(x) = k(x)$.

$x =$ ..... **[3]**

**18** Functions $h$ and $k$ are defined as follows.

$h(x) = \dfrac{4}{x-3}$ for $x \neq 3$

$k(x) = 2x + 1$

**(a)** Find $k(h(5))$.

..... **[2]**

**(b)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

**(c)** Solve the equation $h(k(x)) = 2$.

$x =$ ..... **[3]**

**19** The functions $f$, $g$ and $h$ are defined as follows:

$f(x) = 2x + 3$, $g(x) = x^2 - 4$, $h(x) = 5^x$

**(a)** Find $h(-1)$ as a fraction in its simplest form.

..... **[1]**

**(b)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

**(c)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

**(d)** Find the value of $x$ when $h(x) = 125$.

$x =$ ..... **[1]**

**20** Two functions, $f$ and $g$, are defined as follows:

$$f(x) = \frac{5}{x-2} \qquad g(x) = x + 2$$

**(a)** State the value of $x$ that must be excluded from the domain of $f(x)$.

$x =$ ..... **[1]**

**(b)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation $f(x) = g(x)$.

$x =$ ..... , $x =$ ..... **[4]**

**21** $f(x) = 2x - 3$, $g(x) = x^2 - x - 7$

**(a)** Work out $g(-2)$.

..... **[2]**

**(b)** Find $gf(x)$ in its simplest form.

..... **[3]**

**(c)** Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[3]**

**22** Let $f(x) = \dfrac{2x+1}{3}$ and $g(x) = \dfrac{5}{x-2}$, where $x \neq 2$.

**(a)** Find $gf(x)$.
Give your answer as a single fraction in its simplest form.

..... **[3]**

**(b)** Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[3]**

**(c)** Solve the equation

$$f(x) = g(3)$$

$x =$ ..... **[3]**

**23** $f(x) = 2x - 1$
$g(x) = x^2 - 4$
$h(x) = \dfrac{2x+1}{x-3}, \quad x \neq 3$

**(a)** Work out $g(-3)$.

..... **[1]**

**(b)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

**(c)** Find $gf(x)$ in the form $ax^2 + bx + c$.

$gf(x) =$ ..... **[3]**

**(d)** Solve the equation $f(x) = g(x)$.

$x =$ ..... , $x =$ ..... **[3]**

**24** The functions $f$, $g$ and $h$ are defined as $f(x) = 4 - 3x$, $g(x) = \dfrac{5}{x-2}$, $x \neq 2$, $h(x) = x^2 - x - 7$.

**(a) (i)** Find the value of $f(0.5)$.

..... **[1]**

**(ii)** Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[2]**

**(b)** Find the range of $h(x)$.

..... **[2]**

**(c)** Solve the equation.

$h(x) = 2x - 1$ Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**25** Functions $f$, $g$ and $h$ are defined as follows.

$f(x) = 3x - 2$

$g(x) = x^2 - 4x$

$h(x) = \dfrac{2x+1}{x-1}$ for $x \neq 1$

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

**(b)** Find an expression for $g(f(x))$ in the form $ax^2 + bx + c$.

..... **[3]**

**(c)** Solve the equation $f(g(x)) = x$.
Give your answers correct to 2 decimal places.
You must show all your working.

$x =$ ..... , x =

..... **[4]**

**26** Let $f(x) = 3x + 2$, $g(x) = x^2 - x$ and $h(x) = 5^x$.

**(a)** Find $fg(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[2]**

**(b)** Find $gf(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[3]**

**(c)** Solve the equation

$$gf(x) = 0$$

$x =$ ..... , $x =$ ..... **[3]**

**(d)** Find the value of $x$ when $h(x) = \frac{1}{125}$.

$x =$ ..... **[2]**

**27** $f(x) = \dfrac{3}{x+2}, \quad x \neq -2$
$g(x) = 5x - 4$

**(a)** Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[2]**

**(b)** Find $fg(x)$ as a single fraction in its simplest form.

$fg(x) =$ ..... **[2]**

**(c)** Find $ff(x)$ as a single fraction in its simplest form.

$ff(x) =$ ..... **[3]**

**(d)** Solve the equation $f(x) = g(x)$.
Show all your working.

$x =$ ..... , $x =$ ..... **[4]**

**28** Let $f(x) = \dfrac{2x+5}{x-3}$ for $x \neq 3$, and $g(x) = 3x - 1$.

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

**(b)** Express $fg(x)$ as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

**(c)** Solve the equation $gf(x) = x$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[5]**

**29** Consider the functions $f(x) = \dfrac{3}{x+2}$, $g(x) = (3x+5)^2$ and $h(x) = \dfrac{3x+5}{x-2}$.

**(a)** Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

**(b)** Find $fg(x)$.
Give your answer as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

**(c)** Solve the equation.

$h(x) = x$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , or $x =$ ..... **[5]**

**30** The functions $f$ and $g$ are defined as:

$$f(x) = \frac{2x+3}{x-1} \qquad g(x) = x^2 - 4x$$

**(a)** Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

**(b)** Find $ff(x)$, giving your answer as a single fraction in its simplest form.

$ff(x) =$ ..... **[4]**

**(c)** Solve the equation $g(x) = f(2)$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

**1** **(a)** $g(3) = 3^2 + 1 = 10$ **[M1]**

$f(10) = 2(10) - 5 = 15$ **[A1]**

**(b)** $y = 2x - 5$ OR $x = 2y - 5$ **[M1]**

$\frac{x+5}{2}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

**2** **(a)** 10 **[B1]**

**(b)** 5 **[B1]**

**(c)** $y = 3x - 2 \implies y + 2 = 3x$ **[M1]**

$\frac{x+2}{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

**3** **(a)** **(i)** $-14$ **[B1]**

**(ii)** $f(2) = 3(2) - 2 = 4$ **[M1]**

20 **[A1]**

**(b)** $y = 3x - 2 \implies y + 2 = 3x$ **[M1]**

$\frac{x+2}{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**4** **(a)** $-13$ **[B1]**

**(b)** $y = 4x - 5$ or $x = 4y - 5$ **[M1]**

$\frac{x+5}{4}$ **[A1]**

**(c)** $4\left(\frac{x}{3} + 2\right) - 5$ **[M1]**

$\frac{4x}{3} + 3$ or $\frac{4x+9}{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**5** **(a)** 18 **[B1]**

**(b)** $y = 5x - 2$ or $x = \frac{y+2}{5}$ **[M1]**

$\frac{x+2}{5}$ **[A1]**

**(c)** $g(2) = 2^2 + 3 = 7$ **[M1]**

$f(7) = 5(7) - 2 = 33$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**6** **(a)** 10 **[B1]**

**(b)** $y = 3x - 2 \implies 3x = y + 2$ **[M1]**

$\dfrac{x+2}{3}$ **[A1]**

**(c)** $f(2) = 3(2) - 2 = 4$ **[M1]**

$g(4) = 4^2 + 4 = 20$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**7** **(a)** 13 **[B1]**

**(b)** Finds $f(2) = 1$ **[M1]**

$-2$ **[A1]**

**(c)** Sets $y = 5 - 2x$ and rearranges to $2x = 5 - y$ **[M1]**

$\dfrac{5-x}{2}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**8** **(a)** 5 **[B1]**

**(b)** $y = 4x - 3 \implies x = \dfrac{y+3}{4}$ or $x = 4y - 3$ **[M1]**

$\dfrac{x+3}{4}$ **[A1]**

**(c)** $f(1) = 4(1) - 3 = 1$ **[M1]**

$g(1) = 2 - 5(1) = -3$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**9** **(a)** Calculates $g(3) = 2(3)^2 + 1 = 19$ **[B1]**

Substitutes their 19 into $f(x)$: $4(19) - 3 = 73$ **[B1]**

**(b)** Let $y = 4x - 3$ and attempts to rearrange for $x$ (e.g., $y + 3 = 4x$) **[M1]**

$\dfrac{x+3}{4}$ **[A1]**

**(c)** Sets $4x - 3 = 17$ **[M1]**

$x = 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**10** **(a)** 10 **[B1]**

**(b)** Sets $y = 3x - 2$ or $x = 3y - 2$ OR clearly shows inverse operations $(x + 2) \div 3$ **[M1]**

$\dfrac{x+2}{3}$ **[A1]**

**(c)** Substitutes $f(x)$ into $g(x)$: $(3x-2)^2 + 1$ **[M1]**

Correct expansion of $(3x-2)^2$: $9x^2 - 6x - 6x + 4$ OR $9x^2 - 12x + 4$ **[M1]**

$9x^2 - 12x + 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**11 (a)** $y(x-1) = 3x + 2$ **[M1]**

$xy - 3x = y + 2$ or $x(y-3) = y + 2$ **[M1]**

$\dfrac{x+2}{x-3}$ **[A1]**

**(b)** $\dfrac{3x+2}{x-1} = x - 2$ **[M1]**

$3x + 2 = (x-1)(x-2)$ leading to $3x + 2 = x^2 - 3x + 2$ **[M1]**

$x^2 - 6x = 0$ or $x(x-6) = 0$ **[M1]**

0 and 6 **[A1]**

[Back to Question] *[Total Marks: 7]*

**12 (a)** 1 **[B1]**

**(b)** $y = 2x + 3 \implies 2x = y - 3$ **[M1]**

$\dfrac{x-3}{2}$ **[A1]**

**(c)** $hk(x) = \dfrac{12}{(2x+3)-1}$ **[M1]**

$\dfrac{12}{2x+2} = \dfrac{6}{x+1}$ **[A1]**

**(d)** $k(2) = 2(2) + 3 = 7$ **[B1]**

$\dfrac{12}{x-1} = 7 \implies 12 = 7(x-1)$ **[M1]**

$12 = 7x - 7 \implies 19 = 7x \implies x = \dfrac{19}{7}$ (or $2\dfrac{5}{7}$) **[A1]**

[Back to Question] *[Total Marks: 8]*

**13 (a)** Sets $y = \dfrac{4}{2x-3}$ and multiplies by denominator to obtain $y(2x-3) = 4$ **[M1]**

Expands and rearranges to collect terms in $x$: $2xy = 4 + 3y$ **[M1]**

$\dfrac{4+3x}{2x}$ **[A1]**

**(b)** Finds $g(-3) = 2$ **[B1]**

Equates and attempts to solve: $\frac{4}{2x-3} = 2 \implies 4x - 6 = 4$ **[M1]**

2.5 (or $\frac{5}{2}$) **[A1]**

**(c)** Substitutes $f(x)$ into $g(x)$: $\frac{4}{2x-3} + 5$ **[M1]**

Expresses over a common denominator: $\frac{4 + 5(2x-3)}{2x-3}$ **[M1]**

$\frac{10x - 11}{2x - 3}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

**14** **(a)** $y = \frac{x+3}{x-1} \implies xy - y = x + 3$ **[M1]**

$xy - x = y + 3 \implies x(y-1) = y + 3$ **[M1]**

$\frac{x+3}{x-1}$ **[A1]**

**(b)** $\frac{\frac{x+3}{x-1} + 3}{\frac{x+3}{x-1} - 1}$ **[M1]**

$\frac{\frac{x+3+3(x-1)}{x-1}}{\frac{x+3-(x-1)}{x-1}}$ (or equivalent correct manipulation using a common denominator) **[M1]**

$\frac{x+3+3x-3}{x+3-x+1} = \frac{4x}{4} = x$ with no mathematical errors seen **[A1]**

**(c)** $f(3) = \frac{3+3}{3-1} = 3$ **[B1]**

$2x^2 - 5 = 3$ **[M1]**

$2x^2 = 8 \implies x^2 = 4$ **[M1]**

$2, -2$ **[A1]**

[Back to Question] *[Total Marks: 10]*

**15** **(a)** $y(x-1) = 3x + 2$ or $x(y-1) = 3y + 2$ **[M1]**

$xy - 3x = y + 2$ or $x(y-3) = y + 2$ **[M1]**

$\frac{x+2}{x-3}$ **[A1]**

**(b)** $fg(x) = \frac{3\left(\frac{2}{x+1}\right) + 2}{\frac{2}{x+1} - 1}$ **[M1]**

Numerator becomes $\frac{6 + 2(x+1)}{x+1}$ or Denominator becomes $\frac{2 - (x+1)}{x+1}$ **[M1]**

$\dfrac{2x+8}{1-x}$ **[A1]**

**(c)** $\dfrac{3x+2}{x-1} = \dfrac{x+2}{x-3} \implies (3x+2)(x-3) = (x+2)(x-1)$ **[M1]**

$3x^2 - 7x - 6 = x^2 + x - 2$ **[M1]**

$2x^2 - 8x - 4 = 0$ or $x^2 - 4x - 2 = 0$ **[M1]**

$2 \pm \sqrt{6}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

**16** **(a)** $y(x-1) = 3x + 5$ or $xy - y = 3x + 5$ **[M1]**

$xy - 3x = y + 5$ **[M1]**

$x(y-3) = y + 5$ **[M1]**

$\dfrac{x+5}{x-3}$ **[A1]**

**(b)** $\dfrac{3x+5}{x-1} = x$ **[M1]**

$3x + 5 = x^2 - x$ **[M1]**

$x^2 - 4x - 5 = 0$ **[M1]**

$5$ and $-1$ **[A1]**

**(c)** $f(2) = \dfrac{3(2)+5}{2-1} = 11$ **[M1]**

$11^2 + 1 = 122$ **[A1]**

[Back to Question] *[Total Marks: 10]*

**17** **(a)** $y = \dfrac{2x+1}{3} \implies 3y = 2x + 1$ **[M1]**

$\dfrac{3x-1}{2}$ **[A1]**

**(b)** $hk(x) = \dfrac{2(5-x)+1}{3}$ **[M1]**

$\dfrac{11-2x}{3}$ **[A1]**

**(c)** $\dfrac{2x+1}{3} = 5 - x$ **[M1]**

$2x + 1 = 15 - 3x \implies 5x = 14$ **[M1]**

$2.8$ (or $\dfrac{14}{5}$) **[A1]**

[Back to Question] *[Total Marks: 7]*

**18** **(a)** $h(5) = \dfrac{4}{5-3} = 2$ **[M1]**

$k(2) = 2(2) + 1 = 5$ **[A1]**

**(b)** $y(x-3) = 4$ OR $x(y-3) = 4$ **[M1]**

$\frac{4}{x} + 3$ OR $\frac{4+3x}{x}$ **[A1]**

**(c)** $h(2x+1) = \frac{4}{(2x+1)-3}$ **[M1]**

$\frac{4}{2x-2} = 2 \implies 4 = 4x - 4$ **[M1]**

$x = 2$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**19** **(a)** $\frac{1}{5}$ **[B1]**

**(b)** $(2x+3)^2 - 4$ **[M1]**

$4x^2 + 6x + 6x + 9 - 4$ **[M1]**

$4x^2 + 12x + 5$ **[A1]**

**(c)** $y = 2x + 3 \implies y - 3 = 2x$ **[M1]**

$\frac{x-3}{2}$ **[A1]**

**(d)** 3 **[B1]**

[Back to Question] *[Total Marks: 7]*

**20** **(a)** 2 **[B1]**

**(b)** Sets $y = \frac{5}{x-2}$ and multiplies by $(x-2)$: $y(x-2) = 5$ **[M1]**

Rearranges to isolate $x$: $x - 2 = \frac{5}{y} \implies x = \frac{5}{y} + 2$ OR $xy - 2y = 5 \implies x = \frac{5+2y}{y}$ **[M1]**

$\frac{5}{x} + 2$ OR $\frac{5+2x}{x}$ **[A1]**

**(c)** Equates functions: $\frac{5}{x-2} = x + 2$ **[M1]**

Multiplies by $(x-2)$ to form a quadratic: $5 = (x+2)(x-2)$ **[M1]**

Expands and simplifies to $x^2 = 9$ or $x^2 - 9 = 0$ **[A1]**

3 and $-3$ **[A1]**

[Back to Question] *[Total Marks: 8]*

**21** **(a)** $(-2)^2 - (-2) - 7$ **[M1]**

$-1$ **[A1]**

**(b)** $(2x-3)^2 - (2x-3) - 7$ **[M1]**

$4x^2 - 12x + 9 - 2x + 3 - 7$ **[M1]**

$4x^2 - 14x + 5$ **[A1]**

**(c)** $2x - 3 = x^2 - x - 7$ **[M1]**

$x^2 - 3x - 4 = 0$ or $(x-4)(x+1) = 0$ **[M1]**

$x = 4$ and $x = -1$ **[A1]**

[Back to Question] *[Total Marks: 8]*

**22 (a)** Substitution of $f(x)$ into $g(x)$: $\dfrac{5}{\frac{2x+1}{3} - 2}$ **[M1]**

Writing the denominator as a single fraction: $\dfrac{2x+1-6}{3}$ or $\dfrac{2x-5}{3}$ **[M1]**

$\dfrac{15}{2x-5}$ **[A1]**

**(b)** Let $y = \dfrac{5}{x-2}$ and attempting to multiply by $(x-2)$ or swapping $x$ and $y$ **[M1]**

$x - 2 = \dfrac{5}{y}$ or $xy - 2y = 5$ **[M1]**

$\dfrac{5}{x} + 2$ or $\dfrac{5+2x}{x}$ **[A1]**

**(c)** Evaluates $g(3) = 5$ **[B1]**

Equating $f(x)$ to their $g(3)$: $\dfrac{2x+1}{3} = 5$ **[M1]**

$x = 7$ **[A1]**

[Back to Question] *[Total Marks: 9]*

**23 (a)** 5 **[B1]**

**(b)** $y(x-3) = 2x + 1$ **[M1]**

$xy - 3y = 2x + 1$
$xy - 2x = 3y + 1$
$x(y-2) = 3y + 1$ **[M1]**

$\dfrac{3x+1}{x-2}$ **[A1]**

**(c)** $(2x-1)^2 - 4$ **[M1]**

$4x^2 - 2x - 2x + 1 - 4$ **[M1]**

$4x^2 - 4x - 3$ **[A1]**

**(d)** $2x - 1 = x^2 - 4$
$x^2 - 2x - 3 = 0$ **[M1]**

$(x-3)(x+1) = 0$ **[M1]**

$x = 3, x = -1$ **[A1]**

[Back to Question] *[Total Marks: 10]*

**24 (a) (i)** 2.5 **[B1]**

**(ii)** $g(4-3x) = \dfrac{5}{(4-3x)-2}$ **[M1]**

$\dfrac{5}{2-3x}$ **[A1]**

**(b)** Minimum $x$-coordinate is $-\dfrac{-1}{2(1)} = 0.5$ or via completing square $(x-0.5)^2 - 7.25$ **[M1]**

$h(x) \geq -7.25$ **[A1]**

**(c)** $x^2 - x - 7 = 2x - 1$ leading to $x^2 - 3x - 6 = 0$ **[M1]**

$\dfrac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(-6)}}{2(1)}$ **[M2]**

If M2 not scored, allow M1 for $\dfrac{3 \pm \sqrt{33}}{2}$ or correct use of formula with one error **[M1]**

$-1.37$ and $4.37$ **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**25** **(a)** $y(x-1) = 2x+1 \implies xy - y = 2x + 1$ **[M1]**

$xy - 2x = y + 1 \implies x(y-2) = y + 1$ **[M1]**

$\dfrac{x+1}{x-2}$ **[A1]**

**(b)** $g(3x-2) = (3x-2)^2 - 4(3x-2)$ **[M1]**

$9x^2 - 12x + 4 - 12x + 8$ **[M1]**

$9x^2 - 24x + 12$ **[A1]**

**(c)** $3(x^2 - 4x) - 2 = x$ **[M1]**

$3x^2 - 13x - 2 = 0$ **[A1]**

$x = \dfrac{13 \pm \sqrt{(-13)^2 - 4(3)(-2)}}{2(3)}$ **[M1]**

$x = 4.48$ or $x = -0.15$ **[A1]**

**[Back to Question]** *[Total Marks: 10]*

**26** **(a)** Substitution of $g(x)$ into $f(x)$: $3(x^2 - x) + 2$ **[M1]**

$3x^2 - 3x + 2$ **[A1]**

**(b)** Substitution of $f(x)$ into $g(x)$: $(3x+2)^2 - (3x+2)$ **[M1]**

Correct expansion of $(3x+2)^2$ to $9x^2 + 12x + 4$ **[M1]**

$9x^2 + 9x + 2$ **[A1]**

**(c)** Sets their three-term quadratic $gf(x)$ equal to 0: $9x^2 + 9x + 2 = 0$ **[M1]**

Valid method to solve quadratic (e.g., $(3x+1)(3x+2) = 0$ or correct use of formula) **[M1]**

$x = -\dfrac{1}{3}$ and $x = -\dfrac{2}{3}$ **[A1]**

**(d)** $5^x = 5^{-3}$ **[M1]**

$x = -3$ **[A1]**

[Back to Question] *[Total Marks: 10]*

**27** **(a)** $x = 5y - 4$ or $y = 5x - 4$ **[M1]**

$\frac{x+4}{5}$ **[A1]**

**(b)** $\frac{3}{(5x-4)+2}$ **[M1]**

$\frac{3}{5x-2}$ **[A1]**

**(c)** $\frac{3}{\frac{3}{x+2}+2}$ **[M1]**

$\frac{3}{\frac{3+2(x+2)}{x+2}}$ **[M1]**

$\frac{3(x+2)}{2x+7}$ **[A1]**

**(d)** $\frac{3}{x+2} = 5x - 4$

$3 = (5x-4)(x+2)$ **[M1]**

$3 = 5x^2 + 10x - 4x - 8$

$5x^2 + 6x - 11 = 0$ **[M1]**

$(5x+11)(x-1) = 0$ or correct use of quadratic formula **[M1]**

$x = 1, x = -2.2$ (or $-\frac{11}{5}$) **[A1]**

[Back to Question] *[Total Marks: 11]*

**28** **(a)** Sets $y = \frac{2x+5}{x-3}$ and rearranges to get $y(x-3) = 2x+5$ **[M1]**

Expands and factorises to collect terms in $x$: $xy - 2x = 3y + 5 \implies x(y-2) = 3y+5$ **[M1]**

$\frac{3x+5}{x-2}$ **[A1]**

**(b)** Substitutes $g(x)$ into $f(x)$: $\frac{2(3x-1)+5}{(3x-1)-3}$ **[M1]**

Expands and simplifies the numerator to $6x+3$ **[M1]**

$\frac{6x+3}{3x-4}$ **[A1]**

**(c)** Sets up the equation $3\left(\frac{2x+5}{x-3}\right) - 1 = x$ **[M1]**

Eliminates the denominator to obtain $6x + 15 - (x-3) = x(x-3)$ (or equivalent) **[M1]**

Simplifies to form the correct 3-term quadratic equation: $x^2 - 8x - 18 = 0$ **[A1]**

Correct use of the quadratic formula $x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(-18)}}{2(1)}$ **[M1]**

9.83 and $-1.83$ **[A1]**

[Back to Question] *[Total Marks: 11]*

**29** **(a)** $y = \dfrac{3x+5}{x-2} \implies y(x-2) = 3x+5$ **[M1]**

$xy - 2y = 3x + 5 \implies xy - 3x = 2y + 5 \implies x(y-3) = 2y+5$ **[M1]**

$\dfrac{2x+5}{x-3}$ **[A1]**

**(b)** $\dfrac{3}{(3x+5)^2+2}$ **[M1]**

$\dfrac{3}{9x^2+30x+25+2} = \dfrac{3}{9x^2+30x+27}$ **[M1]**

$\dfrac{3}{3(3x^2+10x+9)} = \dfrac{1}{3x^2+10x+9}$ **[A1]**

**(c)** $\dfrac{3x+5}{x-2} = x \implies 3x+5 = x(x-2)$ **[M1]**

$3x + 5 = x^2 - 2x \implies x^2 - 5x - 5 = 0$ **[A1]**

$\dfrac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(-5)}}{2(1)}$ **[M1]**

$\dfrac{5 \pm \sqrt{25+20}}{2} = \dfrac{5 \pm \sqrt{45}}{2}$ **[M1]**

5.85 and $-0.85$ **[A1]**

[Back to Question] *[Total Marks: 11]*

**30** **(a)** Sets $y = \dfrac{2x+3}{x-1}$ and multiplies: $y(x-1) = 2x+3$ **[M1]**

Expands bracket and collects $x$ terms on one side: $xy - 2x = y + 3$ **[M1]**

Factorises $x$: $x(y-2) = y+3$ **[M1]**

$\dfrac{x+3}{x-2}$ **[A1]**

**(b)** Substitutes $f(x)$ into itself: $\dfrac{2\left(\dfrac{2x+3}{x-1}\right)+3}{\left(\dfrac{2x+3}{x-1}\right)-1}$ **[M1]**

Forms common denominators for numerator and denominator: $\dfrac{\dfrac{2(2x+3)+3(x-1)}{x-1}}{\dfrac{(2x+3)-(x-1)}{x-1}}$ **[M1]**

Correctly simplified numerator: $4x + 6 + 3x - 3 = 7x + 3$ OR correctly simplified denominator: $2x + 3 - x + 1 = x + 4$ **[B1]**

$\dfrac{7x+3}{x+4}$ **[A1]**

**(c)** Calculates $f(2) = \dfrac{2(2)+3}{2-1} = 7$ **[B1]**

Forms correct quadratic equation: $x^2 - 4x - 7 = 0$ **[B1]**

Correct substitution into quadratic formula: $\frac{4 \pm \sqrt{(-4)^2 - 4(1)(-7)}}{2}$ OR $\frac{4 \pm \sqrt{44}}{2}$ **[M1]**

5.32 and $-1.32$ **[A1]**

**[Back to Question]** *[Total Marks: 12]*